

Calculus 3

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Double Integrals - Review of the Definite Integral

If $f(x)$ is defined for $a \leq x \leq b$, we start by dividing the interval $[a, b]$ into n sub-intervals $[x_{i-1}, x_i]$ of equal width $\Delta x = (b - a)/n$ and we choose sample points x_i^* in these sub-intervals. Then we form the Riemann sum

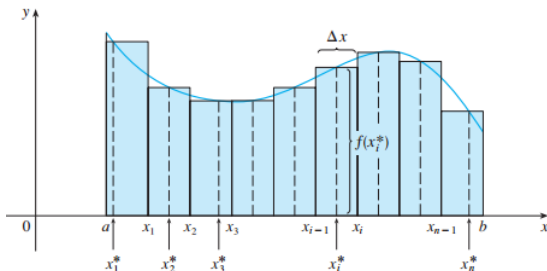
$$\sum_{i=1}^n f(x_i^*) \Delta x$$

and take the limit of such sums as $n \rightarrow \infty$ to obtain the definite integral of f from a to b :

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

Double Integrals - Review of the Definite Integral

In the special case where $f(x) \geq 0$, the Riemann sum can be interpreted as the sum of the areas of the approximating rectangles in the following figure



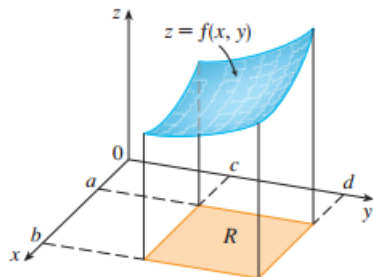
and $\int_a^b f(x)dx$ represents the area under the curve $y = f(x)$ from a to b .

Volumes and Double Integrals

We consider a function f of two variables defined on a closed rectangle

$$R = [a, b] \times [c, d] = \left\{ (x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d \right\}.$$

and we first suppose that $f(x, y) \geq 0$. The graph of f is a surface with equation $z = f(x, y)$.



Let S be the solid that lies above R and under the graph of f , that is

$$S = \left\{ (x, y, z) \in \mathbb{R}^3 \mid 0 \leq z \leq f(x, y), (x, y) \in R \right\}.$$

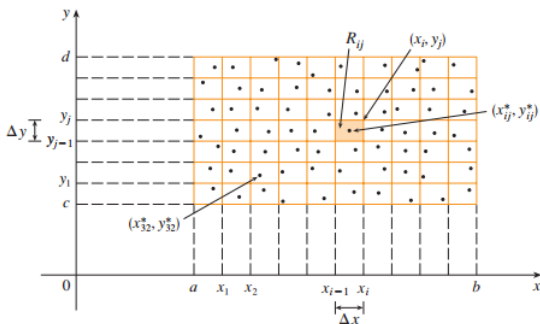
Our goal is to find the volume of S .

Volumes and Double Integrals

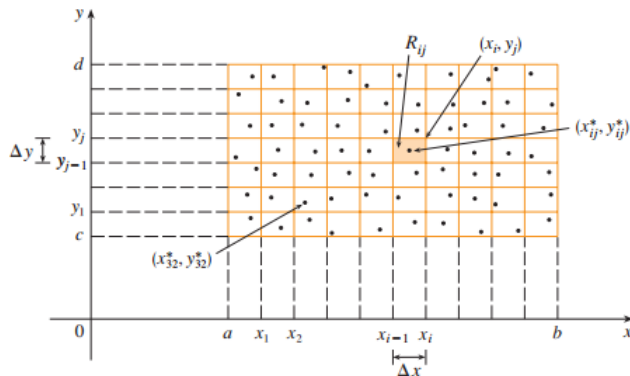
The first step is to divide the rectangle R into sub-rectangles. We accomplish this by

- Dividing the interval $[a, b]$ into m sub-intervals $[x_{i-1}, x_i]$ of equal width $\Delta x = (b - a)/m$.
- Dividing the interval $[c, d]$ into n sub-intervals $[y_{j-1}, y_j]$ of equal width $\Delta y = (d - c)/n$.

By drawing lines parallel to the coordinate axes through the endpoints of these sub-intervals, as in the following figure



Volumes and Double Integrals



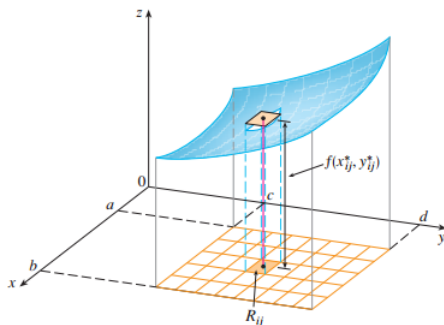
We form the sub-rectangles

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j] = \left\{ (x, y) \in \mathbb{R}^2 \mid x_{i-1} \leq x \leq x_i, y_{j-1} \leq y \leq y_j \right\}$$

each with area $\Delta A = \Delta x \Delta y$.

Volumes and Double Integrals

If we choose a **sample point** (x_{ij}^*, y_{ij}^*) in each R_{ij} , then we can approximate the part of S that lies above each R_{ij} by a thin rectangular box (or “column”) with base R_{ij} and height $f(x_{ij}^*, y_{ij}^*)$ as shown in the following figure



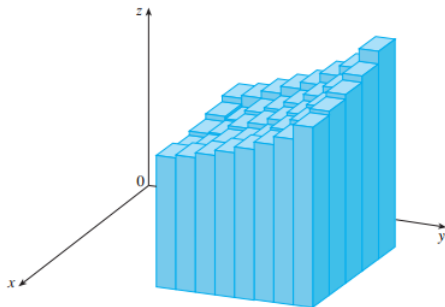
The volume of this box is the height of the box times the area of the base rectangle:

$$f(x_{ij}^*, y_{ij}^*) \Delta A.$$

Volumes and Double Integrals

If we follow this procedure for all the rectangles and add the volumes of the corresponding boxes, we get an approximation to the total volume of S :

$$V \approx \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$



Volumes and Double Integrals

Definition

The **double integral** of f over the rectangle R is

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$

Remark: If $f(x, y) \geq 0$, then the volume V of the solid that lies above the rectangle R and below the surface $z = f(x, y)$ is

$$V = \iint_R f(x, y) dA.$$

Volumes and Double Integrals

Properties of double integrals

a) $\iint_R [f(x, y) \pm g(x, y)] dA = \iint_R f(x, y) dA \pm \iint_R g(x, y) dA.$

b) $\iint_R cf(x, y) dA = c \iint_R f(x, y) dA$ with c is a constant.

c) If $f(x, y) \geq g(x, y)$ for all $(x, y) \in R$, then

$$\iint_R f(x, y) dA \geq \iint_R g(x, y) dA.$$

d) If $R = R_1 \cup R_2$, where R_1 and R_2 don't overlap except perhaps on their boundaries, then

$$\iint_R f(x, y) dA = \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA.$$

e) If we integrate the constant function $f(x, y) = 1$ over a region D , we get the area of D

$$\iint_D 1 dA = A(D).$$

Iterated Integrals

Fubini's theorem

If f is continuous on the rectangle

$$R = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d\}$$

then

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

Remark:

$$\int_a^b \int_c^d f(x, y) dy dx = \int_a^b \left(\int_c^d f(x, y) dy \right) dx$$

$\Rightarrow$ We first integrate with respect to y (holding x fixed) from $y = c$ to $y = d$ and then we integrate the resulting function of x with respect to x from $x = a$ to $x = b$.

$$\int_c^d \int_a^b f(x, y) dx dy = \int_c^d \left(\int_a^b f(x, y) dx \right) dy$$

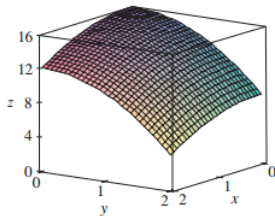
$\Rightarrow$ We first integrate with respect to x (holding y fixed) from $x = a$ to $x = b$ and then we integrate the resulting function of y with respect to y from $y = c$ to $y = d$.

Example: Evaluate the double integral $\iint_R (x - 3y^2) dA$, where

$$R = \left\{ (x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 2, 1 \leq y \leq 2 \right\}.$$

Iterated Integrals

Example: Find the volume of the solid S that is bounded by the elliptic paraboloid $x^2 + 2y^2 + z = 16$, the planes $x = 2$ and $y = 2$, and the three coordinate planes.



Iterated Integrals

Properties of double integrals

Suppose that $f(x, y) = g(x)h(y)$ and $R = [a, b] \times [c, d]$. Then, Fubini's Theorem gives

$$\iint_R f(x, y) dA = \iint_R g(x)h(y) dA = \left(\int_a^b g(x) dx \right) \times \left(\int_c^d h(y) dy \right).$$

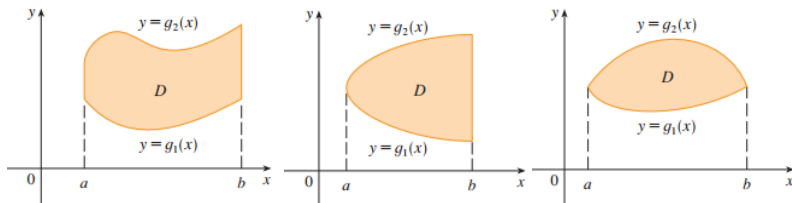
Example: Evaluate the double integral $\iint_R \sin(x) \cos(y) dA$, where $R = [0, \pi/2] \times [0, \pi/2]$.

Double Integrals over General Regions

A plane region D is said to be of **type I** if it lies between the graphs of two continuous functions of x , that is,

$$D = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

where g_1 and g_2 are continuous on $[a, b]$.



Type I

If f is continuous on a type I region

$D = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$ then

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

Double Integrals over General Regions

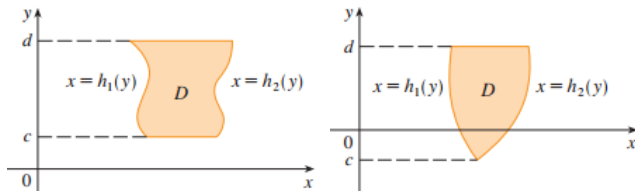
Example: Evaluate $\iint_D (x + 2y) dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.

Double Integrals over General Regions

We also consider plane regions of **type II**, which can be expressed as

$$D = \{(x, y) \in \mathbb{R}^2 \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

where h_1 and h_2 are continuous on $[c, d]$.



Type II

If f is continuous on a type II region

$D = \{(x, y) \in \mathbb{R}^2 \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$ then

$$\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

Double Integrals over General Regions

Example: Evaluate $\iint_D xy \, dA$, where D is the region bounded by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

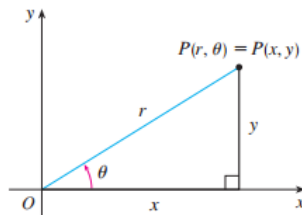
Double Integrals in Polar Coordinates

For each point $P(x, y)$ in Descartes coordinates, we introduce

$$r = OP = \sqrt{x^2 + y^2}, \quad \theta = \angle(\vec{i}, \overrightarrow{OP})$$

then

$$\begin{cases} x = r \cos(\theta) \\ y = r \sin(\theta) \end{cases}$$



$\Rightarrow (r, \theta)$ is the Polar coordinates.

Double Integrals in Polar Coordinates

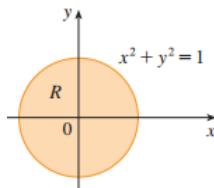
A set D in Descartes coordinates

$$D = \left\{ (x, y) \in \mathbb{R}^2 \mid x \text{ and } y \text{ satisfy the property } T \right\}$$

can be rewritten in the Polar coordinates as form

$$D = \left\{ (r, \theta) \in \mathbb{R}^2 \mid r \text{ and } \theta \text{ satisfy the "similarity" with } T \right\}.$$

Example: A region R in the following figure



can be write as the forms

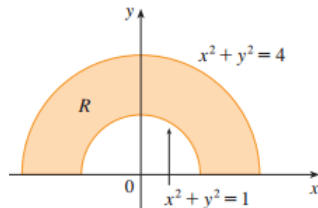
$$R = \left\{ (x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1 \right\}$$

$$R = \left\{ (x, y) \in \mathbb{R}^2 \mid x = r \cos(\theta), y = r \sin(\theta), (r, \theta) \in [0, 1] \times [0, 2\pi] \right\}$$

$$R = \left\{ (r, \theta) \in \mathbb{R}^2 \mid 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi \right\}$$

Double Integrals in Polar Coordinates

Example: A region R in the following figure



can be write as the forms

$$R = \left\{ (x, y) \in \mathbb{R}^2 \mid 1 \leq x^2 + y^2 \leq 4, y \geq 0 \right\}$$

$$R = \left\{ (x, y) \in \mathbb{R}^2 \mid x = r \cos(\theta), y = r \sin(\theta), (r, \theta) \in [1, 2] \times [0, \pi] \right\}$$

$$R = \left\{ (r, \theta) \in \mathbb{R}^2 \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi \right\}$$

Double Integrals in Polar Coordinates

Change to Polar Coordinates in a Double Integral

If f is continuous on a polar rectangle R given by

$$\begin{cases} 0 \leq a \leq r \leq b \\ \alpha \leq \theta \leq \beta \end{cases}$$

Then,

$$\begin{aligned} \iint_R f(x, y) dA &= \int_{\alpha}^{\beta} \int_a^b f(r \cos(\theta), r \sin(\theta)) r dr d\theta \\ &= \int_a^b \int_{\alpha}^{\beta} f(r \cos(\theta), r \sin(\theta)) r d\theta dr \end{aligned}$$

Double Integrals in Polar Coordinates

Example: Evaluate $\iint_R (3x + 4y^2) \, dA$, where R is the region in the upper half-plane bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

Double Integrals in Polar Coordinates

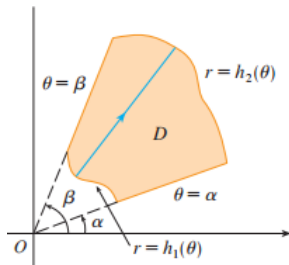
In general case

If f is continuous on a polar region of the form

$$D = \left\{ (r, \theta) \in \mathbb{R}^2 \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta) \right\}$$

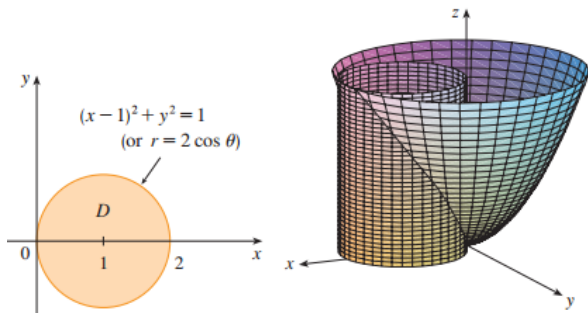
then

$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos(\theta), r \sin(\theta)) r dr d\theta.$$



Double Integrals in Polar Coordinates

Example: Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ above the xy -plane, and inside the cylinder $x^2 + y^2 = 2x$.



Triple Integrals

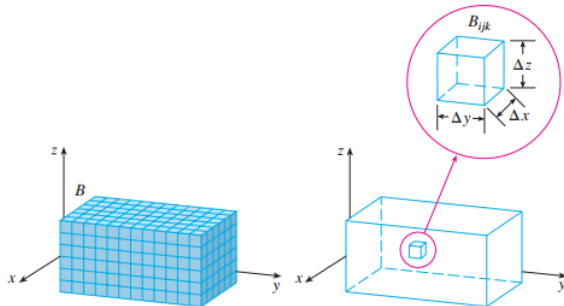
We define

- Single integrals for functions of one variable.
- Double integrals for functions of two variables.
- Triple integrals for functions of three variables.

Let's first deal with the simplest case where f is defined on a rectangular box

$$B = \left\{ (x, y, z) \in \mathbb{R}^3 \mid a \leq x \leq b, c \leq y \leq d, r \leq z \leq s \right\}.$$

Triple Integrals



The first step is to divide B into sub-boxes. We do this by

- Dividing the interval $[a, b]$ into ℓ sub-intervals $[x_{i-1}, x_i]$ of equal width Δx .
- Dividing the interval $[c, d]$ into m sub-intervals $[y_{j-1}, y_j]$ of equal width Δy .
- Dividing the interval $[r, s]$ into n sub-intervals $[z_{k-1}, z_k]$ of equal width Δz .

The planes through the endpoints of these sub-intervals parallel to the coordinate planes divide the box B into $\ell \times m \times n$ sub-boxes

$$B_{ijk} = [x_{i-1}, x_i] \times [y_{j-1}, y_j] \times [z_{k-1}, z_k].$$

Each sub-box has volume $\Delta V = \Delta x \Delta y \Delta z$.

Triple Integrals

Then we form the **triple Riemann sum**

$$\sum_{i=1}^{\ell} \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V$$

where the sample point $(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)$ is in B_{ijk} .

Definition

The **triple integral** of f over the box B is

$$\iiint_B f(x, y, z) dV = \lim_{\ell, m, n \rightarrow \infty} \sum_{i=1}^{\ell} \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V$$

if this limit exists.

Fubini's Theorem for Triple Integrals

If f is continuous on the rectangular box $B = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint_B f(x, y, z) dV = \int_r^s \int_c^d \int_a^b f(x, y, z) dx dy dz.$$

Example: Evaluate the triple integral $\iiint_B xyz^2 dV$, where B is the rectangular box given by

$$B = \{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq 1, -1 \leq y \leq 2, 0 \leq z \leq 3\}.$$

Triple Integrals

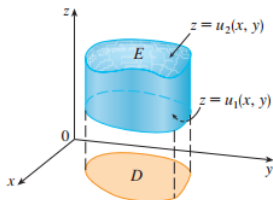
Type I

A solid region E is said to be of **type 1** if it lies between the graphs of two continuous functions of x and y , that is,

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y) \right\}$$

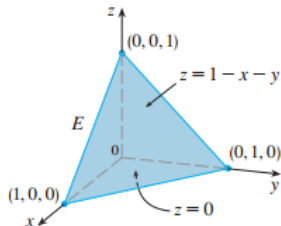
where D is the projection of E onto the xy -plane. Then,

$$\iiint_E f(x, y, z) dV = \iint_D \left(\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz \right) dA. \quad (1)$$



Triple Integrals

Example: Evaluate $\iiint_E z \, dV$, where E is the solid tetrahedron bounded by the four planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$.



Triple Integrals

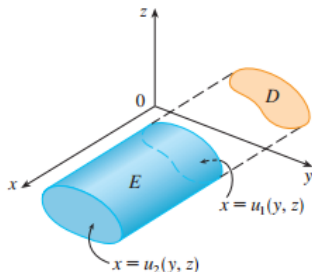
Type 2

A solid region E is of **type 2** if it is of the form

$$E = \{(x, y, z) \in \mathbb{R}^3 \mid (y, z) \in D, u_1(y, z) \leq x \leq u_2(y, z)\}$$

where D is the projection of E onto the yz -plane. Then,

$$\iiint_E f(x, y, z) dV = \iint_D \left(\int_{u_1(y, z)}^{u_2(y, z)} f(x, y, z) dx \right) dA$$



Triple Integrals

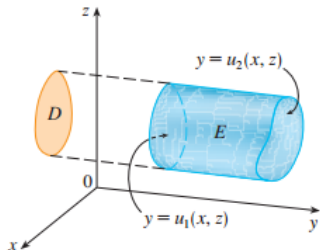
Type 3

A solid region E is of **type 3** if it is of the form

$$E = \{(x, y, z) \in \mathbb{R}^3 \mid (x, z) \in D, u_1(x, z) \leq y \leq u_2(x, z)\}$$

where D is the projection of E onto the xz -plane. Then,

$$\iiint_E f(x, y, z) dV = \iint_D \left(\int_{u_1(x, z)}^{u_2(x, z)} f(x, y, z) dy \right) dA$$



Triple Integrals

Example: Evaluate $\iiint_E \sqrt{x^2 + z^2} dV$, where E is the region bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$.

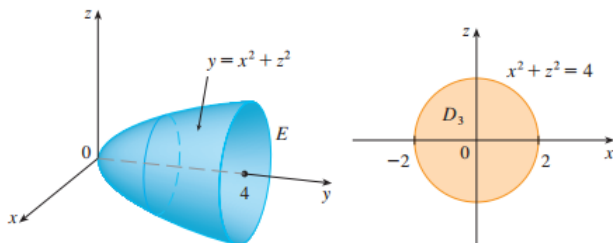


Figure: Region of integration (the left) and Projection onto xz -plane (the right).

Triple Integrals

The volume of E

The triple integral does represent the volume of E :

$$V(E) = \iiint_E dV.$$

Example: Use a triple integral to find the volume of the tetrahedron T bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$ and $z = 0$.

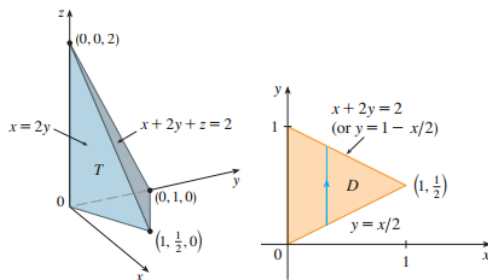


Figure: The tetrahedron T (the left) and its projection D onto xy -plane (the right).

Triple Integrals in Cylindrical Coordinates

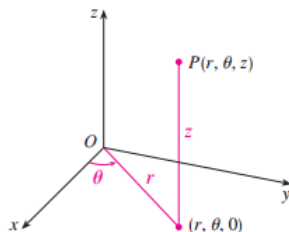


Figure: The Cylindrical Coordinates of a point.

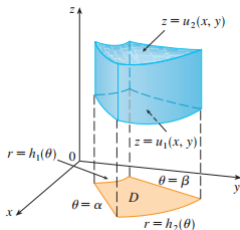
In the **cylindrical coordinate** system, a point P in three-dimensional space is represented by the ordered triple (r, θ, z) , where r and θ are polar coordinates of the projection of P onto the xy -plane and z is the directed distance from the xy -plane to P .

To convert from cylindrical to rectangular coordinates, we use the equations

$$x = r \cos(\theta), \quad y = r \sin(\theta), \quad z = z.$$

Evaluating Triple Integrals with Cylindrical Coordinates

Suppose that E is a type 1 region whose projection D onto the xy -plane is conveniently described in polar coordinates



In particular, suppose that f is continuous and

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y) \right\}$$

where D is given in polar coordinates by

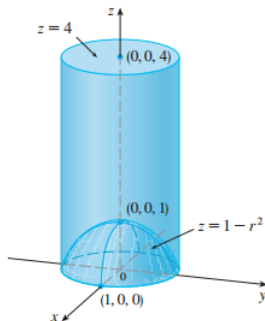
$$D = \left\{ (r, \theta) \in \mathbb{R}^2 \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta) \right\}.$$

Then,

$$\iiint_E f(x, y, z) dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos(\theta), r \sin(\theta))}^{u_2(r \cos(\theta), r \sin(\theta))} f(r \cos(\theta), r \sin(\theta), z) r dz dr d\theta.$$

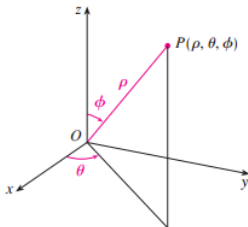
Evaluating Triple Integrals with Cylindrical Coordinates

Example: Evaluate $\iiint_E \sqrt{x^2 + y^2} dV$, where E lies within the cylinder $x^2 + y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$.



Triple Integrals in spherical coordinates

The **spherical coordinates** (ρ, θ, ϕ) of a point P in space are shown in the following figure



where

- $\rho = |OP|$ is the distance from the origin to P .
- θ is the same angle as in cylindrical coordinates.
- ϕ is the angle between the positive z -axis and the line segment OP .

Note that: $\rho \geq 0$ and $0 \leq \phi \leq \pi$.

Triple Integrals in spherical coordinates

To convert from spherical to rectangular coordinates, we use the equations

$$\begin{cases} x = \rho \sin(\phi) \cos(\theta) \\ y = \rho \sin(\phi) \sin(\theta) \\ z = \rho \cos(\phi) \end{cases}$$

Evaluating Triple Integrals with Spherical Coordinates

Spherical wedge

If E is a spherical wedge given by

$$E = \{(\rho, \theta, \phi) \in \mathbb{R}^3 \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$$

then

$$\begin{aligned} & \iiint_E f(x, y, z) dV \\ &= \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)) \rho^2 \sin(\phi) d\rho d\theta d\phi. \end{aligned}$$

Example: Evaluate $\iiint_E e^{(x^2+y^2+z^2)^{3/2}} dV$, where V is the unit ball
 $V = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 1\}.$

Evaluating Triple Integrals with Spherical Coordinates

Spherical wedge

If E is a spherical wedge given by

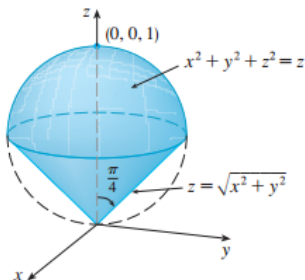
$$E = \left\{ (\rho, \theta, \phi) \in \mathbb{R}^3 \mid \alpha \leq \theta \leq \beta, c \leq \phi \leq d, g_1(\theta, \phi) \leq \rho \leq g_2(\theta, \phi) \right\}$$

then

$$\begin{aligned} & \iiint_E f(x, y, z) dV \\ &= \int_{g_1(\theta, \phi)}^{g_2(\theta, \phi)} \int_c^d \int_\alpha^\beta f(\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)) \rho^2 \sin(\phi) d\theta d\phi d\rho. \end{aligned}$$

Evaluating Triple Integrals with Spherical Coordinates

Example: Use spherical coordinates to find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$.



THANK YOU FOR YOUR
ATTENTION !